

Agent-Guided Hierarchical Semantic Search over 3D Scene Records

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Abstract. Photorealistic 3D digital twins remain difficult to query because visual observations, objects, and spatial evidence grow faster than the context available to a reasoning agent. We study a representation-agnostic hierarchy that prunes a scene before semantic grounding. The system supports deterministic lexical and embedding controls, an instruction-tuned language agent, an original Cursor-Agent/MCP construction path, and a hierarchy constructed directly from posed RGB-D observations. On five indoor/outdoor captures (97 RGB-D views; 150 queries), a complete Qwen2.5 run checks 9.63 views per query and obtains hit@1 / hit@3 of 0.368 / 0.696 on 125 verified positives while recording every prompt, call, native token count, latency, and trace. Five direct Cursor MCP calls construct scene partitions without access to manual zones and reach pairwise F1 0.616 against the overlapping manual reference. A separate CLIP constructor reads zero manual ViewJSON records and reaches pairwise F1 0.487 from RGB, depth, and poses. In the deterministic control, mean per-query view reduction is 75.5%, while the hit@1 difference is inconclusive and hit@3 declines significantly under a paired bootstrap. ScanNet relation-aware graph search reaches Acc@0.25 0.604 versus 0.500 without relations while checking 90.3% fewer objects than flat search. Replica is reported only as an oracle-map sanity ceiling.

Keywords: Digital twins · 3D scene graphs · RGB-D · Language agents · Grounding.

1 Introduction

Digital twins based on neural rendering and 3D Gaussian Splatting (3DGS) [9] support immersive inspection, but appearance alone does not answer questions such as “where can I sit near reception?” or “which view shows the stage screen?” A practical interface must ground language in scene evidence while controlling how much of a large map reaches the semantic reasoner.

Flat retrieval is a strong baseline: it exposes every record to one scorer and avoids pruning mistakes. Its cost, however, grows linearly with the map. Scene graphs and language fields provide rich semantics [4, 10, 12, 14], yet two questions remain important for digital twins: (i) can a hierarchy reduce records and tokens without silently changing the scorer, and (ii) can the hierarchy itself be obtained from raw capture data rather than hand-authored semantic records?

We answer both questions with a reproducible prototype. It separates construction cost from per-query cost and separates automatic predictions from manual evaluation references. The same 150-query benchmark is executed by deterministic controls, a local instruction-tuned language agent, and a frozen vision-language model (VLM). All agent prompts, model calls, exact tokenizer counts, response tokens, latencies, branch decisions, and ranked views are saved. The original agent architecture is evaluated independently: Cursor invokes the hierarchy-construction MCP tool once per scene and semantically partitions its geometric candidates without seeing manual zones. A stricter automatic constructor consumes only RGB images, depth maps, and camera poses; the manual hierarchy is opened only after the automatic artifact is written.

Contributions.

- A typed, traceable hierarchy for top-down semantic search over shared 3D-scene records, with lexical, embedding, and instruction-agent scorers.
- A complete 150-query instruction-agent evaluation with native token and call accounting, rather than a stub-only demonstration.
- A measured Cursor-Agent/MCP construction path with pose and pose-plus-semantic controls, plus a raw RGB-D-to-hierarchy pipeline using frozen CLIP, depth statistics, poses, and predicted branch labels. Both are evaluated against but not constructed from manual zone references.
- Same-input controls on five captures, BBQ-aligned Replica/ScanNet object-grounding pilots, and native ConceptGraphs/LangSplat feasibility runs, with protocol differences stated explicitly.

2 Related Work

Language-aligned 3D representations. LERF [10], LangSplat [14], Semantic Gaussians [5], and LEGS [20] attach language or semantic features to neural fields and Gaussian primitives. ConceptFusion [7] and OpenScene [13] construct open-vocabulary 3D representations. Our contribution is orthogonal: we control the records presented at query time and can operate over a manual index, an oracle object map, or a predicted map.

Scene graphs and hierarchical reasoning. ConceptGraphs [4], HOV-SG [19], and Hydra [6] organize entities and relations for mapping and planning. BBQ [12] is the closest object-grounding reference: it combines a 3D object graph with deductive language reasoning and reports $\text{Acc}@k$ on Replica and ScanNet. SayPlan [17] grounds an LLM in a hierarchical 3D scene graph for scalable planning, while Search3D [18] studies hierarchical open-vocabulary 3D segmentation. We focus on controlled branch pruning, complete accounting, and the manual-versus-automatic hierarchy gap.

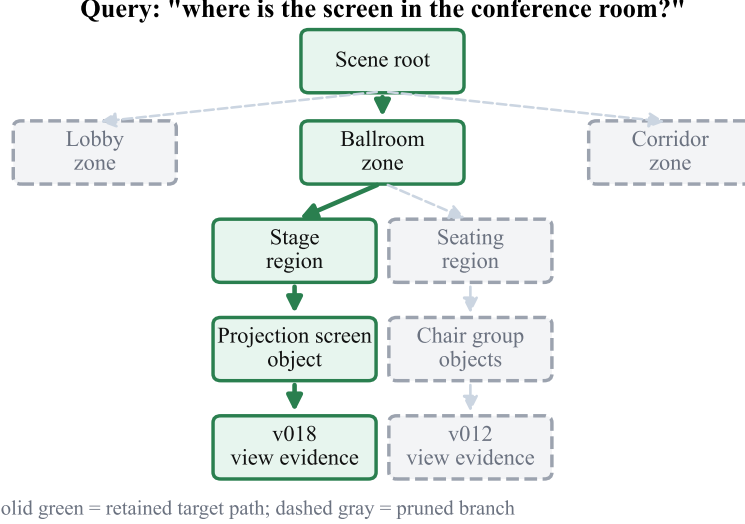


Fig. 1: Top-down query execution. A root-to-branch decision produces a candidate subset; the same leaf scorer used by flat search ranks the retained views or objects. Every visited node and retained ID is written to the trace.

Grounding and context control. ScanRefer [1] established language grounding in RGB-D scans, and OpenLex3D [8] advocates tiered evaluation for open-vocabulary 3D representations. Context-pruning work in language systems similarly separates relevance selection from generation [2, 3, 11]. Our evaluation therefore reports retrieval quality and efficiency jointly; fewer records alone do not establish a better method.

3 Method

3.1 Hierarchical scene records

Let a scene hierarchy be $H = (V, E)$ with root v_0 . Each node is

$$v_i = (\text{id}_i, \tau_i, A_i, x_i, R_i, B_i, I_i). \quad (1)$$

It stores a type τ_i , its ancestor path A_i , searchable content x_i , spatial relations R_i , optional 2D/3D boxes B_i , and supporting observations I_i . Branch nodes represent zones or regions; entity nodes represent objects, landmarks, signs, or facilities; evidence nodes point to RGB-D views. Containment edges and typed spatial relations remain separate.

This representation supports a controlled comparison. Flat search scores all eligible leaves. Hierarchical search first selects branches and then applies the identical leaf scorer. If N is the full leaf set and N_q the retained set, flat scoring is $O(N)$ and branch-pruned leaf scoring is $O(N_q)$ after a small top-level gate. An explicit fallback may restore the full set; it is recorded rather than hidden.

3.2 Agent/MCP hierarchy construction

The semantic-record construction track implements the original two-pass agent architecture. Given camera translations p_i , the MCP server first builds an adaptive proximity graph. Its edge threshold is

$$t = 3.5 \operatorname{median}_i \min_{j \neq i} \|p_i - p_j\|_2, \quad (2)$$

and connected components form conservative geometric candidates. A deterministic semantic control merges components only when their view summaries contain either a complementary facing pair or a shared room type and landmark.

For the agent variant, Cursor calls `build_spatial_clusters` once for each scene with geometry assistance enabled, then partitions and labels the returned candidates from the saved room, facing, landmark, and object summaries. It cannot read the manual-zone reference, call the tree persistence tool, or modify scene data. All five decisions and rationales are frozen before evaluation. Cursor’s UI exposes neither provider tokens nor a stable wall-clock field, so both are reported as unavailable rather than inferred. This construction experiment is separate from the query-side Qwen benchmark: Cursor measures MCP orchestration, while the pinned local model provides reproducible query metrics and exact tokens.

3.3 Deterministic and embedding controls

The lexical control normalizes case, stop words, aliases, and plurals. A branch score is token overlap between the query and concatenated branch content. Branches with at least half the maximum positive overlap are retained; if all branch scores are zero, every branch remains eligible. Leaves are ranked by overlap, normalized overlap, node-type priority, and stable ID. The flat variant applies the same ranking to every leaf.

The semantic control replaces overlap with BGE-small-en-v1.5 cosine similarity. Query, branch, and leaf text use the model’s native tokenizer. Both controls are deterministic and credential-independent; they establish whether gains come from hierarchy or from a changed scorer.

3.4 Instruction-agent traversal

We execute Qwen2.5-0.5B-Instruct [15] locally in deterministic CPU inference. For a query q and candidate c , the agent receives:

```
Judge scene-query relevance using semantic meaning and
affordances. Answer exactly Yes or No.
Query: <q>. Candidate: <c>. Relevant? Answer exactly Yes or No.
```

At the first phase, candidates are top-level branches. The two branches with largest normalized next-token probability $p(\text{Yes} \mid q, c)$ are retained. The second phase applies the same instruction to their view records and returns the top

three. Candidate prompts are batched within a phase, but a phase remains one recorded model invocation. Generation is greedy and emits exactly one decision token per candidate.

Each query trace stores the system message, every rendered chat prompt, candidate-to-alias mapping, **Yes/No** probabilities, generated decisions, ranked IDs, branch IDs, native input/output tokens, model calls, and wall-clock latency. This is an actual instruction-model execution; it is not a lexical stub or an unverified provider cache.

3.5 Automatic raw RGB-D hierarchy

The automatic constructor has a strict source boundary: it may read `transforms.json`, RGB images, and depth arrays, but not ViewJSON, manual zone files, or the checked-in tree. For each view i , it computes a frozen CLIP image embedding f_i , camera translation p_i , and depth descriptor d_i containing valid-depth ratio and five quantiles. Standardized features are concatenated as follows. We use the ViT-B/32 CLIP checkpoint [16].

$$z_i = [f_i; 0.5 p_i; 0.2 d_i]. \quad (3)$$

Deterministic farthest-point initialization and k-means cluster the views. The number of clusters maximizes silhouette score over $k = 2, \dots, \min(8, \lfloor n/3 \rfloor)$. A fixed scene-agnostic vocabulary of 20 place categories is encoded with the same CLIP model; each cluster receives the category whose text embedding has highest cosine similarity to its mean image embedding.

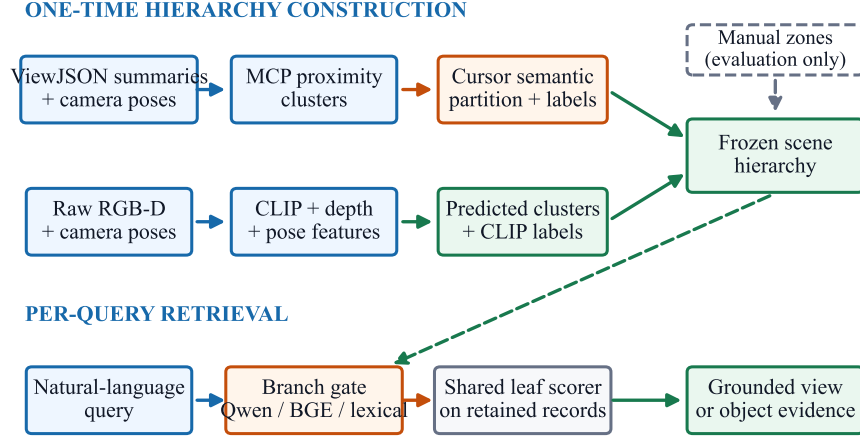
The raw hierarchy is serialized before the evaluator opens manual zone references. Structural agreement is measured by pairwise precision, recall, F1, and Rand index over co-clustered view pairs. For query evaluation, one CLIP text embedding ranks (i) all images, (ii) images in the two best manual branches, or (iii) images in the two best automatic branches. Thus all three variants share the image/text scorer.

3.6 Object grounding and spatial relations

For public object maps, a parsed query produces a target, optional relation, and optional anchor. Exact labels, lexical overlap, type, and deterministic affordance expansions form the base object score. Relations such as *near*, left/right, above/below, and front/behind use stored links or box centers. A satisfied relation adds 0.35, a resolved violation subtracts 0.35×0.35 , and unresolved geometry contributes zero. A fallback expands the universe when confidence is below 0.45, geometry is invalid, or a required relation remains unresolved. Exact source object IDs and boxes are preserved for Recall@1 and 3D IoU.

3.7 Accounting

Construction and query costs are never combined. Construction reports source files, model invocations, text tokens, images processed, and latency. Each query



Construction and query costs are reported separately; Cursor UI tokens are unavailable, never imputed.

Fig. 2: Two complementary construction tracks feed the same query pipeline. Cursor semantically partitions MCP proximity clusters built from scene records; the raw track bypasses ViewJSON and predicts branches from RGB-D and poses. Manual zones are opened only for evaluation after artifacts are frozen.

reports checked views/objects/nodes, serialized context, native or explicitly estimated tokens, calls, and latency. Provider tokens are not inferred. Local tokenizer counts are exact for the associated model; legacy character-divided-by-four values are labeled estimates.

4 Datasets and Evaluation

4.1 Internal captures

The internal benchmark contains five manually captured indoor/outdoor scenes: Conference Hall, Museum, Theater, outdoor street, and outdoor drone. It has 97 posed RGB-D views, 97 manual ViewJSON records, 1,066 semantic items, and 150 queries. Of these, 125 positive queries have verified expected view IDs and 25 negative queries are excluded from $\text{hit}@k$ because they lack independent negative GT. Query types include simple objects, attributes, affordances, relations, and multi-hop descriptions.

4.2 Public datasets and external executions

The public track follows the BBQ scene subset where available. Replica uses room0–2 and office0–4 (8 scenes, 575 official boxes, 56 queries including 8 negatives). ScanNet uses scenes 0011_00, 0030_00, 0046_00, 0086_00, 0222_00,

Digital-twin pilot scenes (manual captures)



Fig. 3: The five captured environments; the lower-right panel repeats a Conference Hall stage-screen view used in the qualitative demo. Manual records support the controlled reference track, while the automatic track reads only RGB images, depth arrays, and poses.

0378_00, 0389_00, and 0435_00 (392 official boxes). The core ScanNet track has 48 mapped Nr3D/Sr3D+ queries; an extended track has 64 queries, including 8 negative and 8 additional positive cases.

These object-grounding pilots search oracle semantic candidates and therefore measure retrieval over official IDs and boxes, not automatic semantic-map perception. Native ConceptGraphs builds predicted object maps from sampled RGB-D; native LangSplat learns a language field. Their protocols differ and they are reported as execution references, not a ranking.

4.3 Metrics

Efficiency uses mean checked views/objects, input tokens, model calls, and latency. View retrieval uses hit@1, hit@3, and mean reciprocal rank (MRR) on the 125 positive internal queries. Object grounding uses exact-ID Recall@ k , MRR, 3D IoU, and Acc@ τ . Negative accuracy is reported only where verified negative labels exist. Structural hierarchy quality uses pairwise F1 and Rand index. Missing predictions stay in the relevant denominator with score zero.

5 Experiments

5.1 Complete instruction-agent benchmark

Table 1 and Fig. 4 report all 150 queries. The instruction agent reaches hit@1 0.368, hit@3 0.696, and MRR 0.509 on the 125 labeled positives while checking 9.63 views. It uses 2.00 batched phase calls, 1582.9 input tokens, and 16.4 generated decision tokens per query on average. CPU latency is 6.24s/query. Across the run, the saved ledger contains 300 model calls and 239893 exact native tokens.

Table 1: Complete instruction-agent execution. Tokens are native Qwen tokenizer counts; latency is deterministic CPU wall time.

Method	Q	GT	hit@1	hit@3	MRR	Views	Calls	Tokens
Qwen graph agent	150	125	0.368	0.696	0.509	9.63	2.00	1582.9

Complete 150-query instruction-agent evaluation

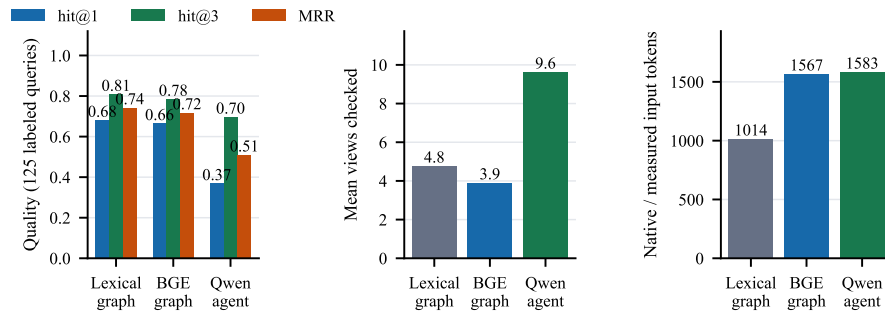


Fig. 4: Instruction-agent evidence against deterministic graph controls. Tokenizer semantics differ by model, so token bars measure each method’s actual input representation rather than a billing comparison.

This run directly answers whether the hierarchy can be traversed by a real language model. The answer is yes operationally: all prompts and responses are valid and reproducible. Quality remains an empirical result rather than a claim of LLM superiority; the smaller Qwen model is included to make the complete run credential-independent.

5.2 Raw RGB-D hierarchy versus manual reference

The constructor processes 97 RGB images, 97 depth maps, and 97 poses using 15 model calls and 151 exact CLIP text tokens for semantic branch labeling. Construction takes 14.15s on CPU and reads zero manual ViewJSON or zone files. Against the manual overlapping zones, macro pairwise F1 is 0.487 and macro Rand index is 0.637.

At query time, flat CLIP obtains hit@1 / hit@3 0.432 / 0.624 while checking 19.40 views. Manual-hierarchy CLIP obtains 0.424 / 0.592 while checking 10.76 views. The automatic hierarchy obtains 0.368 / 0.528 while checking 10.00 views (Fig. 5). This experiment exposes rather than hides the cost of removing manual semantics.

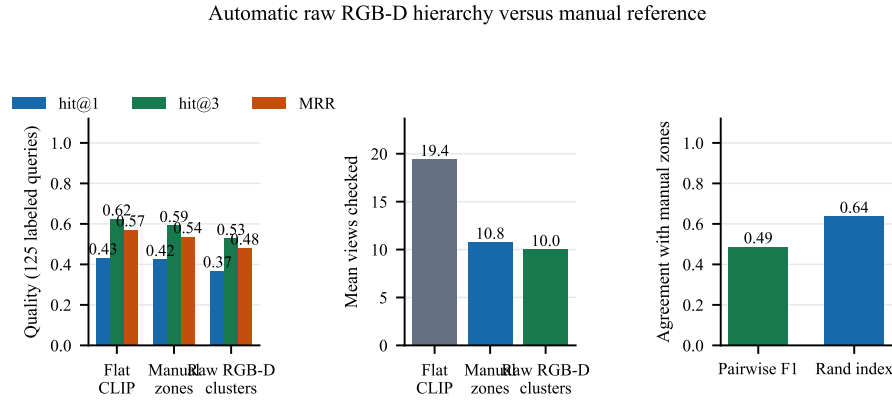


Fig. 5: Automatic hierarchy construction and retrieval. Structure is scored only after the raw artifact is frozen; all retrieval variants use the same CLIP query/image scorer.

5.3 Semantic-record construction variants

Table 2 and Fig. 6 compare four hierarchies under the same 97 records, 150 queries, and fixed leaf scorer. The Cursor-Agent/MCP hierarchy uses 5 direct tool calls, creates 18 zones, and reaches pairwise F1 0.616 and best-zone Jaccard 0.610. It obtains hit@1 / hit@3 0.704 / 0.784 while checking 5.53 views and serializing 1193 estimated query tokens on average.

Pose-plus-semantic construction obtains the strongest retrieval quality (0.784 / 0.904) but is structurally less similar to the overlapping manual zones (pairwise F1 0.570). Cursor is structurally closest among the automatic semantic-record variants, whereas no single construction dominates structure, retrieval quality, and query cost. Relative to the manual reference, Cursor changes hit@1 by +0.024 and hit@3 by −0.024, while checking 5.53 rather than 4.77 views; this is a comparable controlled retrieval profile, not a claim of equivalent independent-GT accuracy.

Table 2: Hierarchy construction from shared semantic records. Quality uses 125 verified positives; efficiency uses all 150 queries. Tokens are measured serialized query context, not Cursor provider billing.

Variant	Zones	F1	Jacc.	hit@1	hit@3	Views	Tokens
Manual reference	26	1.000	1.000	.680	.808	4.77	1,014
Pose only	10	0.588	.526	.760	0.880	6.07	1,780
Pose + semantic	8	0.570	.504	0.784	0.904	5.80	1,747
Cursor Agent + MCP	18	0.616	0.610	0.704	0.784	5.53	1193

Hierarchy construction from shared semantic records

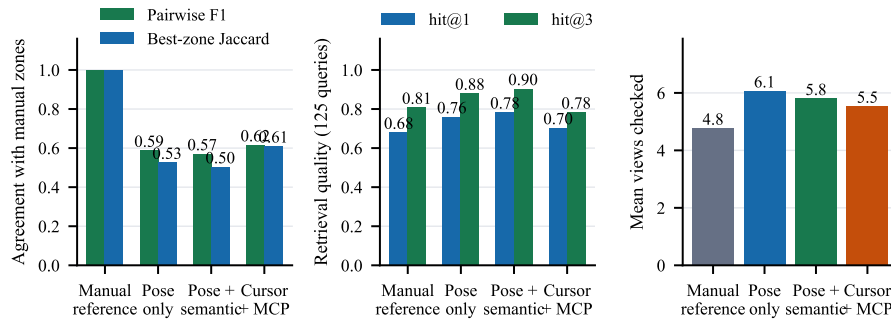


Fig. 6: Construction variants under a fixed query policy. Manual-reference agreement is descriptive rather than independent GT because zones overlap and were authored during the pilot.

5.4 Deterministic same-input controls

On the manual records, flat lexical checks 19.4 views and 3,168 estimated tokens per query. Hierarchical lexical checks 4.77 views and 1,014 tokens. Mean per-query reductions are 75.5% (95% CI [73.2, 77.7]) and 68.2% ([65.3, 71.0]), respectively. The ratio of the displayed mean view counts is separately $1 - 4.77/19.40 = 75.4\%$; it differs slightly because the primary estimate averages each query’s reduction before rounding.

On the 125 positive queries, hit@1 is 0.680 versus 0.768 flat. Its paired graph-minus-flat difference is -0.088 with 95% CI [-0.184, 0.008], which includes zero and is therefore inconclusive. Hit@3 is 0.808 versus 0.928; its difference is -0.120 [-0.184, -0.056], a significant decline under the paired 10,000-resample percentile bootstrap. BGE graph search checks 3.90 views and 1,567 native tokens, with hit@1 0.664 and hit@3 0.784; flat BGE obtains 0.680 and 0.928. These controls show that aggressive pruning saves context but can remove relevant evidence.

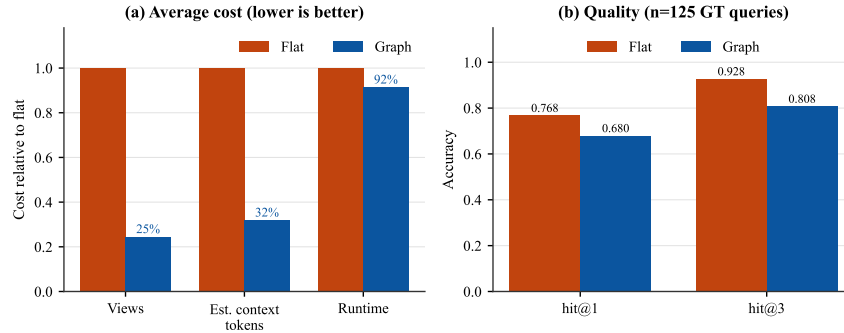


Fig. 7: Manual-record lexical control. The hierarchy reduces checked views and estimated context but does not preserve flat hit@ k .

5.5 Replica and ScanNet object grounding

Table 3 updates the earlier pilots with the calibrated v3 relation-aware scorer. Replica remains an oracle-map ceiling: graph and flat reach exact-ID Recall@1 and Acc@0.25 of 1.0 on 48 positives, and negative accuracy 1.0 on 8 negatives. Graph checks 12.11 objects versus 71.88 flat.

On the 48-query ScanNet core, graph and flat both reach exact-ID Recall@1 / Acc@0.25 of 0.604. Graph Recall@3 is 0.812 and MRR is 0.718 while checking 4.73 objects and 219 estimated tokens, versus 49.0 objects and 2,117 tokens flat. Removing relation scoring reduces Acc@0.25 to 0.500, so the latest spatial reranker helps rather than hurts. On the 64-query extended track, graph Recall@1 is 0.607, Acc@0.25 is 0.589 over 56 box-eligible positives, and negative accuracy is 1.0 over 8 negatives.

The superseded reranker underperformed because it applied low-confidence tie-breaking and frame-dependent directional predicates even when geometry was unresolved. The v3 scorer uses confidence, geometry-validity, and margin gates and abstains from a relation adjustment when those checks fail.

Table 3: Latest BBQ-aligned oracle-map pilots. Efficiency is mean checked objects; tokens are deterministic serialized-context estimates.

Track	Variant	R@1	R@3	Acc@.25	Objects	Tokens
Replica (56; 48+)	graph	1.000	1.000	1.000	12.11	303
	flat lexical	1.000	1.000	1.000	71.88	1,765
ScanNet (48; 48+)	graph	.604	.812	.604	4.73	219
	no relation	.500	.771	.500	22.71	987
	flat lexical	.604	.833	.604	49.00	2,117
ScanNet ext. (64; 56+)	graph	.607	.786	.589	4.38	200

Replica’s exact-match ceiling is only a protocol sanity check and cannot distinguish retrieval quality. On ScanNet, equality at $R@1/Acc@0.25$ is not presented as universal quality preservation: graph $R@3$ is 0.812 versus 0.833 flat, and removing relations lowers $R@1/Acc@0.25$ to 0.500. The public result is therefore a measured efficiency–quality profile under a frozen oracle-map protocol, not evidence that all methods are equivalent.

5.6 Native external executions

ConceptGraphs constructs predicted maps on all eight ScanNet and eight Replica scenes. Its $Acc@0.25$ is 0.0625 on each 48-positive box evaluation; mean 3D IoU is 0.0696 and 0.0624, respectively. LangSplat completes RGB 3DGS optimization, SAM/CLIP preprocessing, autoencoder compression, language-field optimization, rendering, and six-query retrieval on ScanNet 0011_00. It selects an expected view for 4/6 queries but emits points rather than 3D boxes.

Table 4: Native external runs. Inputs and map construction differ from the oracle-map experiments; values do not define a method ranking.

Method	Scenes	Q	Map	Quality	s/query	Tokens
ConceptGraphs–ScanNet	8	48	predicted	$Acc@.25=.063$	.336	627
ConceptGraphs–Replica	8	56	predicted	$Acc@.25=.063$	.274	459
LangSplat	1	6	learned field	view hit=.667	2.037	74

The LangSplat run uses 3,000 RGB and language iterations with SAM-B/CLIP-B on a 6 GB GPU instead of the paper-scale 30,000-iteration/24 GB setting. It is a complete hardware-adapted execution, not a paper-scale reproduction. ConceptGraphs uses sampled frames and predicted boxes. No official BBQ code execution is claimed.

6 Discussion and Limitations

Three findings emerge. First, hierarchy is an effective cost-control mechanism: all graph variants inspect fewer views or objects. Second, retrieval quality depends on both branch construction and branch scoring. The manual lexical and BGE controls lose $hit@k$, the Cursor/MCP experiment shows that the intended agent construction path is executable, and the raw hierarchy quantifies the additional cost of predicting branches without semantic records. Third, a real instruction model can execute the traversal with complete accounting, but the small CPU model should be viewed as a reproducible agent baseline rather than a state-of-the-art reasoning result.

The five captures and manual labels are small and not independent public GT. The automatic constructor predicts coarse view clusters rather than object-instance segmentation. CLIP labels come from a fixed generic vocabulary. Public

oracle-map results isolate retrieval and cannot establish semantic perception accuracy. Native ConceptGraphs and LangSplat runs are diagnostic because compute, frame sampling, geometry, and output types differ. Segmentation mIoU remains unavailable without paired per-point predictions. Finally, no fair official BBQ execution is included, so we make no superiority claim.

7 Reproducibility

The release contains the 150-query list, exact expected-view labels, per-query results, raw traces, construction artifacts, CSV summaries, run configurations, model revisions, and validation reports. The instruction run uses `Qwen/Qwen2.5-0.5B-Instruct` revision `7ae557604adf67be50417f59c2c2f167def9a775`; the raw constructor uses `openai/clip-vit-base-patch32`. Both run in deterministic CPU inference. A validator checks 150 instruction traces, token arithmetic, five direct Cursor/MCP decisions, four semantic-record construction variants, 97 raw RGB-D/pose records, zero manual-construction reads, 150 three-way raw-hierarchy query rows, all denominators, and the paired bootstrap values. Figure and table scripts consume these frozen JSON outputs directly.

8 Conclusion

We presented a traceable hierarchy for semantic search over 3D scene records and evaluated both the querying agent and the origin of the hierarchy. A complete instruction-tuned language model traverses all 150 queries with prompts, calls, tokens, latency, and quality saved. A separate raw RGB-D constructor builds and labels branches without manual ViewJSON, then exposes its structural and retrieval gap relative to the manual hierarchy. The original Cursor-Agent/MCP path is also measured through frozen construction decisions rather than conflated with Qwen’s query metrics. Together with same-input controls, calibrated ScanNet/Replica pilots, and native external executions, these results turn hierarchy from an assumed component into a measured part of the digital-twin search pipeline.

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